

Inverting Conditional Probability

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Introduction

- We know how to compute $\Pr(A \mid B)$ when we know $\Pr(A \wedge B)$ and $\Pr(B)$.
- But often what we're *given* is a set of conditional probabilities, and we need to work out a different conditional probability.
- Today we'll learn how to **invert** conditional probabilities.

Associated Reading

Odds and Ends, Chapter 8

A burglary was committed, and fingerprints resembling mine were found at the scene. Let:

- I = I am innocent
- F = these fingerprints were found

From forensic science, we can estimate $\Pr(F \mid I)$ — the chance that fingerprints like mine appear at a random crime scene where I'm innocent.

But what matters for criminal responsibility is $\Pr(I \mid F)$. These are very different things, and conflating them is a serious error.

The Table Method

A Real-World Problem

- 3% of men have disease D .
- A man you know has no special risk factors.
- There is a test for D : among those with D , 90% test positive; among those without, only 5% test positive.

This man tested **positive**. What is the probability he has disease D ?

condition positive (P)
the number of real positive cases in the data
condition negative (N)
the number of real negative cases in the data
true positive (TP)
exp. with hit
true negative (TN)
exp. with correct rejection
false positive (FP)
exp. with false alarm, Type I error
false negative (FN)
exp. with miss, Type II error
sensitivity, recall, hit rate, or true positive rate (TPR)
$TPR = \frac{TP}{P} = \frac{TP}{TP + FN} = 1 - FNR$
specificity, selectivity or true negative rate (TNR)
$TNR = \frac{TN}{N} = \frac{TN}{TN + FP} = 1 - FPR$
precision or positive predictive value (PPV)
$PPV = \frac{TP}{TP + FP} = 1 - FDR$
negative predictive value (NPV)
$NPV = \frac{TN}{TN + FN} = 1 - FOR$
miss rate or false negative rate (FNR)
$FNR = \frac{FN}{P} = \frac{FN}{FN + TP} = 1 - TPR$
fail-out or false positive rate (FPR)
$FPR = \frac{FP}{N} = \frac{FP}{FP + TN} = 1 - TNR$
false discovery rate (FDR)
$FDR = \frac{FP}{FP + TP} = 1 - PPV$
false omission rate (FOR)
$FOR = \frac{FN}{FN + TN} = 1 - NPV$
Prevalence Threshold (PT)
$PT = \sqrt{\frac{TPR \cdot (TNR + 1) + TNR - 1}{(FPR + TNR - 1)}}$
Threat score (TS) or critical success index (CSI)
$TS = \frac{TP}{TP + FN + FP}$

Figure 1: Testing terminology (sensitivity and specificity)

How to Solve These Problems

Build a table with:

- The possible **states of the world** on the rows (Has Disease / No Disease)
- The possible **test results** on the columns (Positive / Negative)
- Fill each cell with $\Pr(\text{state} \wedge \text{result})$

The key formula used throughout:

$$\Pr(X \wedge Y) = \Pr(X) \times \Pr(Y \mid X)$$

Start with what we know:

- $\Pr(\text{Disease}) = 0.03$, so $\Pr(\text{No Disease}) = 0.97$
- $\Pr(\text{Positive} \mid \text{Disease}) = 0.9$, so $\Pr(\text{Negative} \mid \text{Disease}) = 0.1$
- $\Pr(\text{Positive} \mid \text{No Disease}) = 0.05$, so
 $\Pr(\text{Negative} \mid \text{No Disease}) = 0.95$

$$\Pr(\text{Disease} \wedge \text{Positive}) = 0.03 \times 0.9 = 0.027$$

	Positive	Negative
Disease	0.027	
No Disease		

$$\Pr(\text{Disease} \wedge \text{Negative}) = 0.03 \times 0.1 = 0.003$$

	Positive	Negative
Disease	0.027	0.003
No Disease		

$$\Pr(\text{No Disease} \wedge \text{Positive}) = 0.97 \times 0.05 = 0.0485$$

	Positive	Negative
Disease	0.027	0.003
No Disease	0.0485	

$$\Pr(\text{No Disease} \wedge \text{Negative}) = 0.97 \times 0.95 = 0.9215$$

	Positive	Negative
Disease	0.027	0.003
No Disease	0.0485	0.9215

Checking the Table

	Positive	Negative
Disease	0.027	0.003
No Disease	0.0485	0.9215

- Top row sums to 0.03 $\square$ (that's $\Pr(\text{Disease})$)
- Bottom row sums to 0.97 $\square$ (that's $\Pr(\text{No Disease})$)
- Everything sums to 1 $\square$

We want $\Pr(\text{Disease} \mid \text{Positive})$.

$$\Pr(\text{Disease} \mid \text{Positive}) = \frac{\Pr(\text{Disease} \wedge \text{Positive})}{\Pr(\text{Positive})}$$

- $\Pr(\text{Disease} \wedge \text{Positive}) = 0.027$
- $\Pr(\text{Positive}) = 0.027 + 0.0485 = 0.0755$

$$\Pr(\text{Disease} \mid \text{Positive}) = \frac{0.027}{0.0755} \approx 0.36$$

The Surprising Upshot

- The test looked pretty reliable: 90% sensitivity, 95% specificity.
- But testing positive only raises the probability of disease from 3% to about **36%**.

Why? Because the disease is **rare**. Even though the test is fairly reliable, false positives outnumber true positives when the base rate is low.

This pattern is general. When testing for rare conditions, a positive result is worrying but usually doesn't mean you almost certainly have the disease.

General Strategy

The General A/B Strategy

For any problem where you want $\Pr(A \mid B)$ from information of the form $\Pr(B \mid A)$, $\Pr(A)$, etc.:

1. Make a 2×2 table with $A, \neg A$ on rows and $B, \neg B$ on columns.
2. Fill each cell using $\Pr(X \wedge Y) = \Pr(X \mid Y) \Pr(Y)$.
3. Add up rows to get $\Pr(A)$, $\Pr(\neg A)$; columns to get $\Pr(B)$, $\Pr(\neg B)$.
4. Use the conditional probability formula to find the answer.

Example

Given:

- $\Pr(A) = 0.6$
- $\Pr(B \mid A) = 0.3$
- $\Pr(B \mid \neg A) = 0.8$

Find $\Pr(B)$ and $\Pr(A \mid B)$.

From negation: $\Pr(\neg A) = 0.4$, $\Pr(\neg B \mid A) = 0.7$,
 $\Pr(\neg B \mid \neg A) = 0.2$.

		B	$\neg B$
A	A		
	$\neg A$		

$$\Pr(A \wedge B) = 0.3 \times 0.6 = 0.18$$

$$\Pr(A \wedge \neg B) = 0.7 \times 0.6 = 0.42$$

$$\Pr(\neg A \wedge B) = 0.8 \times 0.4 = 0.32$$

$$\Pr(\neg A \wedge \neg B) = 0.2 \times 0.4 = 0.08$$

	B	$\neg B$
A	0.18	0.42
$\neg A$	0.32	0.08

	B	$\neg B$
A	0.18	0.42
$\neg A$	0.32	0.08

- $\Pr(B) = 0.18 + 0.32 = 0.5$

$$\Pr(A \mid B) = \frac{0.18}{0.5} = 0.36$$

Learning B drops the probability of A from 0.6 to 0.36.

Probability Rules

It helps to see the algebra behind the table method.

The Negation Rule

$$\Pr(\neg A) = 1 - \Pr(A)$$

This holds because A and $\neg A$ are exclusive and exhaustive.

The Multiplication Rule

From the definition of conditional probability:

$$\Pr(A \mid B) = \frac{\Pr(A \wedge B)}{\Pr(B)}$$

Multiply both sides by $\Pr(B)$:

$$\Pr(A \wedge B) = \Pr(A \mid B) \Pr(B)$$

This is what we use to fill in every cell of the table.

B is equivalent to $(A \wedge B) \vee (\neg A \wedge B)$, and those are exclusive, so:

$$\Pr(B) = \Pr(B \mid A) \Pr(A) + \Pr(B \mid \neg A) \Pr(\neg A)$$

This is what lets us compute column totals from the cells.

Combining the multiplication rule and total probability:

$$\Pr(A \mid B) = \frac{\Pr(B \mid A) \Pr(A)}{\Pr(B \mid A) \Pr(A) + \Pr(B \mid \neg A) \Pr(\neg A)}$$

This is **Bayes's Theorem**. It's very famous — but note that it's just a compact way of writing out what the table method does.

$$\Pr(A \mid B) = \frac{\Pr(B \mid A) \Pr(A)}{\Pr(B)}$$

The table method and Bayes's theorem always give the same answer. Use whichever you find easier — for most exam problems, the table is clearer and less error-prone.

Base Rates

A Famous Example

A cab was involved in a hit-and-run at night. Two cab companies operate in the city: Green (85% of cabs) and Blue (15%).

A witness identified the cab as Blue. The court tested the witness under similar conditions: they correctly identified the color of cabs 80% of the time.

What is the probability the cab was Blue?

Setting Up the Table

Let $\Pr(\text{Blue}) = 0.15$, $\Pr(\text{Green}) = 0.85$.

From the reliability data:

- $\Pr(\text{Looks Blue} \mid \text{Is Blue}) = 0.8$
- $\Pr(\text{Looks Blue} \mid \text{Is Green}) = 0.2$

	Looks Blue	Looks Green
Is Blue	$0.15 \times 0.8 = 0.12$	$0.15 \times 0.2 = 0.03$
Is Green	$0.85 \times 0.2 = 0.17$	$0.85 \times 0.8 = 0.68$

$$\Pr(\text{Is Blue} \mid \text{Looks Blue}) = \frac{0.12}{0.12 + 0.17} = \frac{0.12}{0.29} \approx 0.41$$

Even though the witness is 80% accurate, it is still **more likely the cab was green**.

The Lesson: Base Rates

- When one possibility is much more likely *a priori*, it takes strong evidence to overturn it.
- Here, Green cabs outnumber Blue 85:15 — so even a fairly reliable witness can't overcome that.

We often conflate the **direction** of recent evidence with its **strength**. The witness supports “Blue” over “Green” — but the overall balance still favors Green.

A Caution About This Example

The cab example assumes the witness's accuracy is stated in a particular way. It could be read differently:

If “80% accurate” means:

- $\Pr(\text{Correct} \mid \text{Looks Blue}) = 0.8$ (i.e., whenever they say “Blue”, they’re right 80% of the time)

Then the answer would simply be 0.8. English descriptions of accuracy data are often ambiguous. Always check exactly what you are being told.

- Base rates matter enormously.
- “This evidence supports H ” is different from “ H is now probably true.”
- Use the table method to avoid being misled by salient recent evidence.

Multiple Hypotheses

Worked Example: The Urn Problem

An urn contains 4 marbles, each either blue or green. The number of blue marbles is equally likely to be 0, 1, 2, 3, or 4. We draw 3 marbles **with replacement** and observe: **blue, green, blue**.

Question: What is the probability the urn contains exactly 1 blue marble?

Let H_k = the urn contains k blue marbles, for $k = 0, 1, 2, 3, 4$.

Prior: $\Pr(H_k) = \frac{1}{5}$ for each k — we start with no reason to favour any particular composition.

Computing the Likelihoods

Let E = the sequence blue, green, blue. Since draws are **independent** (with replacement), we multiply:

$$\Pr(E \mid H_k) = \frac{k}{4} \times \frac{4-k}{4} \times \frac{k}{4} = \frac{k^2(4-k)}{64}$$

H_k	$k^2(4-k)$	$\Pr(E \mid H_k)$
H_0	0	0
H_1	3	3/64
H_2	8	8/64
H_3	9	9/64
H_4	0	0

Applying Bayes's Theorem

$$\Pr(H_1 | E) = \frac{\Pr(E | H_1) \Pr(H_1)}{\sum_{k=0}^4 \Pr(E | H_k) \Pr(H_k)}$$

Since all priors equal $1/5$, the $1/5$ factors cancel and the denominator is just the sum of the likelihoods (times $1/5$):

$$\Pr(E) = \frac{1}{5} \cdot \frac{0 + 3 + 8 + 9 + 0}{64} = \frac{1}{5} \cdot \frac{20}{64} = \frac{1}{16}$$

$$\Pr(H_1 | E) = \frac{(3/64) \times (1/5)}{1/16} = \frac{3}{320} \times 16 = \frac{3}{20}$$

What the Evidence Tells Us

With a uniform prior, the posterior is **proportional to the likelihood** — the evidence does all the work:

H_k	Likelihood	Posterior
H_0	0	0
H_1	3/64	3/20
H_2	8/64	8/20
H_3	9/64	9/20
H_4	0	0

The most likely urn composition after observing blue-green-blue is H_3 — 3 blue marbles. Intuitively: we saw two blues and one green, which is consistent with a 3:1 ratio. The uniform prior means the evidence speaks for itself.

- We move from probability to **decisions**.
- How should we act when the outcomes of our choices depend on uncertain facts?
- We'll introduce expected utility as the key tool.